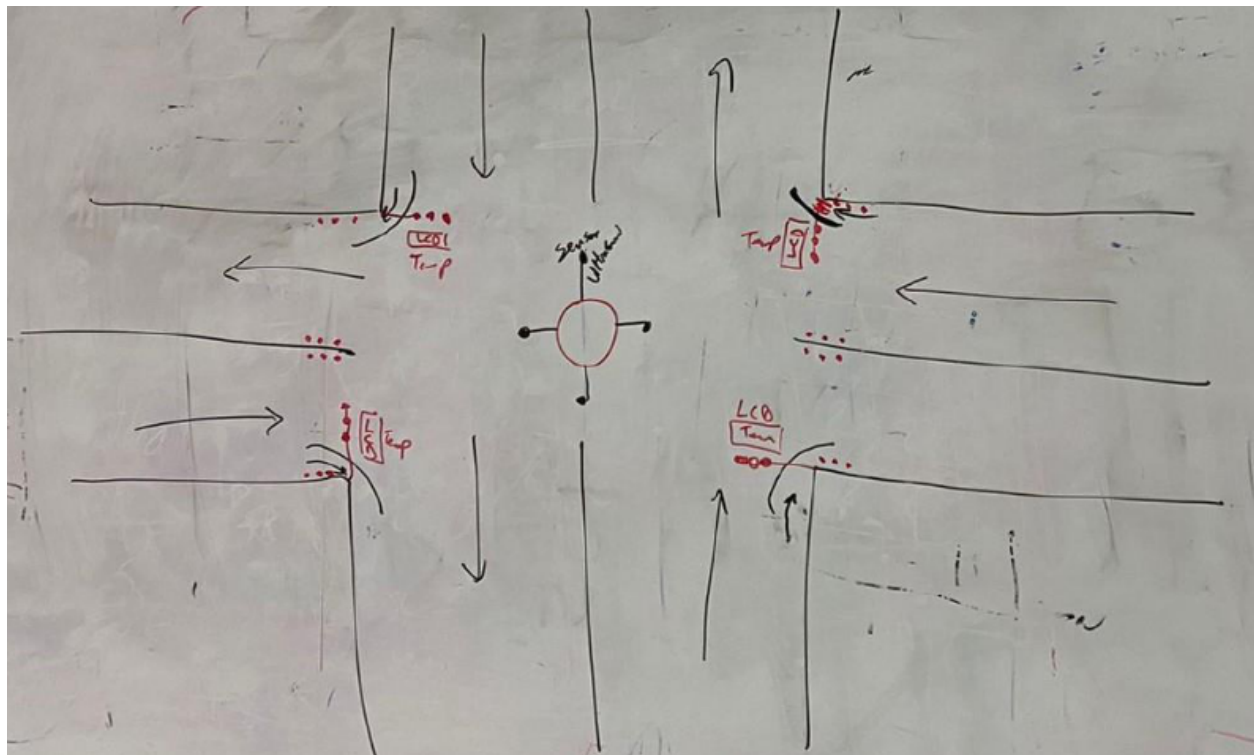


Project Description:

The primary objective of this project is to develop a real advanced traffic signal system (in Egypt or Germany or etc...) for highway intersections, employing the BASYS3 FPGA board to facilitate smart traffic control. Utilizing VHDL programming, the FPGA will serve as the central control unit, managing traffic signals based on external inputs and predefined rules.



We have an intersection cross point of 4 streets having 2 lanes each of 2 different ways.
Each street should have 2 traffic lights (one for the cars and one for pedestrians)

- Specs:
 1. If a car crossed the street while the traffic light is red, a sensor should detect it and an alarm should go off.
 2. The traffic light should measure the temperature and displays it on the 7-segment of the basys3 board.

3. All LEDs of all traffic lights are 220 Volts.
- Bonus:
 1. A camera that captures the car plate number if the car crossed the street, while the traffic light is red.
 2. An extra lane for the cars entering the nearest street like the German system.
 3. LCD to display more information (temperature on LCD not on the 7 segments of the board).
 4. Advertising screen that displays ads.